Artifact Three:

Databases Narrative

The artifact that I chose for my artifact three is my DAT 375 project two. This project was created last term, which was April 2026. This project was to analyze the relationship between property crime in Miami-Dade County and storm events. For the original project I used pandas, Python, matplotlib, and MySQL to help me retrieve, read, analyze, and clean the data. In the project, I filtered the crime records to only include property crime, removing empty and missing values. I also filtered the storm events to only show storm events that occurred in Miami-Dade County. I reformatted the dates to match to be able to make the datasets merge. In my analysis, I used charts to show visualizations of the relationship between crime and storm events. One was a bar chart showing property crime during storm events, and the other was a line chart showing property crimes during the events over time.

The reason I selected this artifact is because I wanted to show my ability in data migration, data management, using Python, querying, data cleaning, and data analysis. This project required me to work with large datasets, answer an important question through proper analysis, and to represent the results through visualizations. I believe this shows my ability to work with datasets, databases, and proper programming tools to solve problems. I improved the artifact by migrating the data from MySQL to SQLite. I created a database with SQLite and imported the original data being crime data and storm data into proper tables and used the proper SQL queries to retrieve the data from the database I created with SQLite. My enhancements maintained the original functionality of the project while improving portability, querying, and security. I also improved the organization and added comments which I did not have in the original version of the project.

I believe I met the course outcomes that I planned to align with during this enhancement. For this artifact, I thought it aligned with two outcomes. Outcome 3 which was to design and evaluate computing solutions using algorithmic principles and computer science practices. I believe I did this through creating a SQLite database, querying the data, cleaning the data, and properly filtering the data. I also believe my enhancements align with outcome four, which was to demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing solutions. I did this through using Python, pandas, matplotlib, SQLite, and querying to read, analyze, manage, process, and visualize the data. I do not have any updates to my outcome-coverage plans. My enhancement aligned with both the outcomes that I originally chose in the first module.

Through my enhancement, it helped strengthen my understanding of databases, data analysis, and database management. I learned how to migrate a project from one database to another database and keep the same workflow. I also learned that while the datasets I was working with are decent in size, the larger the dataset is the more complex the migration can become. When doing the migration, I also needed to learn how to create tables with SQLite and execute queries and how they may differ. One challenge I faced was creating the SQLite tables from the original dataset and modifying the database connection while making sure the workflow remained the same. In the original project, the database was already created in MySQL. When migrating from MySQL to SQLite I need to make sure that the date fields were formatted the same so the datasets could be merged correctly. This was important for importing the mpdcrimedata and the storm_events datasets for the SQLite database schema I created. This made it important to learn to create the database and check that when reading the datasets, I ended up with the same results as the original. This let me know that I created the database correctly but also my querying was correct as well. While maintaining the workflow from the original I continued to test and verify that the filtering, merging, and visualization answered the original question and comparison of the relationship between property crime in Miami-Dade County during certain storm events.